Computer Vision Performance & Comparative Analysis

Object Detection, Motion Estimation, and Depth Estimation Evaluations

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1. Object Detection Performance Comparison

Given detection results for Model A (Viola–Jones) and Model B (YOLO):

• Model A: True Positives (TP) = 80, False Positives (FP) = 20, False Negatives (FN) = 30

• Model B: True Positives (TP) = 90, False Positives (FP) = 15, False Negatives (FN) = 20

(a) Calculation of Precision and Recall:

1. Precision Formula: Precision = TP / (TP + FP)
   - Model A Precision = 80 / (80 + 20) = 80 / 100 = 0.8000 or 80.00%
   - Model B Precision = 90 / (90 + 15) = 90 / 105 ≈ 0.8571 or 85.71%

2. Recall Formula: Recall = TP / (TP + FN)
   - Model A Recall = 80 / (80 + 30) = 80 / 110 ≈ 0.7273 or 72.73%
   - Model B Recall = 90 / (90 + 20) = 90 / 110 ≈ 0.8182 or 81.82%

Model	TP	FP	FN	Precision	Recall
Model A (Viola-Jones)	80	20	30	80.00%	72.73%
Model B (YOLO)	90	15	20	**85.71%**	**81.82%**

(b) Performance Comparison & Justification:

Model B (YOLO) provides significantly better overall object detection performance.

• Higher Precision: Model B achieves 85.71% precision compared to Model A's 80.00%, meaning Model B produces fewer false alarms (False Positives).
• Higher Recall: Model B achieves 81.82% recall compared to Model A's 72.73%, meaning Model B successfully detects a higher proportion of true objects (fewer missed detections / False Negatives).
• Overall Superiority: Model B outperforms Model A across both evaluation metrics simultaneously.

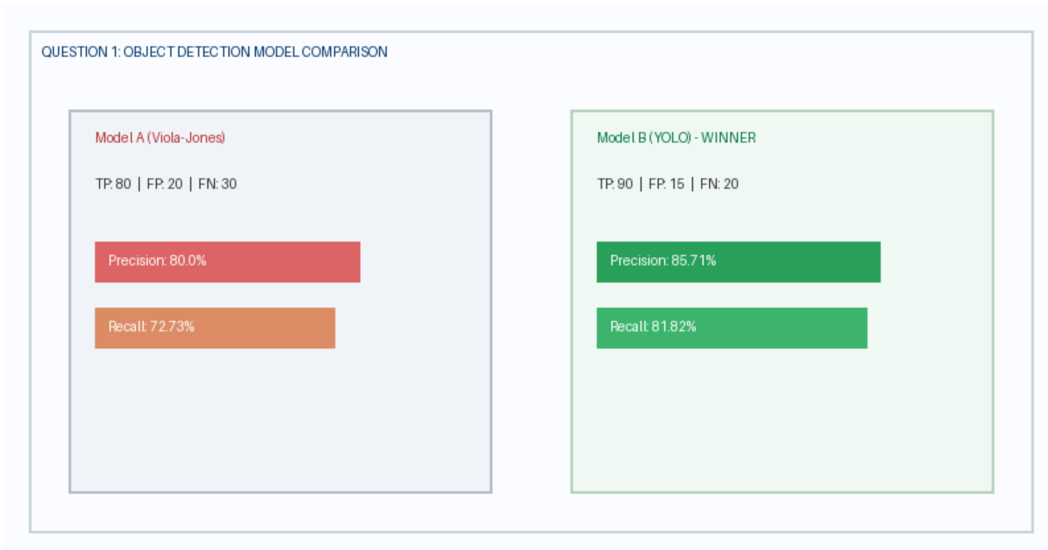

Figure 1: Object detection performance comparison metrics between Model A and Model B.

2. Motion Estimation Error Analysis

Displacement errors in pixels across three consecutive frames:
• Method A displacement errors: 2 px, 3 px, 2 px
• Method B displacement errors: 3 px, 5 px, 4 px

(a) Calculation of Average Displacement Error (ADE):

• Method A Average Error = (2 + 3 + 2) / 3 = 7 / 3 ≈ 2.33 pixels
• Method B Average Error = (3 + 5 + 4) / 3 = 12 / 3 = 4.00 pixels

Method	Frame 1 Error	Frame 2 Error	Frame 3 Error	Total Error	Average Error (ADE)
Method A	2 px	3 px	2 px	7 px	**2.33 pixels**

Method B	3 px	5 px	4 px	12 px	4.00 pixels

(b) & (c) Accuracy Determination & Justification:

Method A provides more accurate motion estimation.

• Justification: Method A has a significantly lower average displacement error (2.33 pixels) compared to Method B (4.00 pixels).

• Frame-by-Frame Consistency: Method A exhibits lower error across every single frame (Frame 1: 2px vs 3px; Frame 2: 3px vs 5px; Frame 3: 2px vs 4px), demonstrating higher stability and motion vector tracking precision.

Figure 2: Motion estimation displacement error trajectory over three consecutive video frames.

3. Depth Estimation Comparison

Depth measurements for three objects:

• Actual Depths: Object 1 = 2.0 m, Object 2 = 3.0 m, Object 3 = 4.0 m

• Stereo Vision Estimates: 2.0 m, 3.0 m, 4.0 m

• Structure-from-Motion (SfM) Estimates: 2.5 m, 3.8 m, 5.0 m

(a) & (b) Absolute Error and Average Absolute Error Calculations:

Absolute Error Formula: | Estimated Depth - Actual Depth |

1. Stereo Vision System Errors:

- Object 1: | 2.0 - 2.0 | = 0.0 m
- Object 2: | 3.0 - 3.0 | = 0.0 m
- Object 3: | 4.0 - 4.0 | = 0.0 m
- Stereo Vision Average Absolute Error (MAE) = (0.0 + 0.0 + 0.0) / 3 = 0.00 m

2. Structure-from-Motion (SfM) System Errors:

- Object 1: | 2.5 - 2.0 | = 0.5 m
- Object 2: | 3.8 - 3.0 | = 0.8 m
- Object 3: | 5.0 - 4.0 | = 1.0 m
- SfM Average Absolute Error (MAE) = (0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 ≈ 0.77 m

Object	Actual Depth	Stereo Est.	Stereo Error	SfM Est.	SfM Error
Object 1	2.0 m	2.0 m	**0.0 m**	2.5 m	0.5 m
Object 2	3.0 m	3.0 m	**0.0 m**	3.8 m	0.8 m
Object 3	4.0 m	4.0 m	**0.0 m**	5.0 m	1.0 m

(c) Identification of Superior Method & Justification:

Stereo Vision provides significantly superior depth estimation performance.

• Perfect Accuracy: Stereo Vision achieves an Average Absolute Error of 0.00 m (exact match to true physical depth for all objects).

• SfM Discrepancy: Structure-from-Motion exhibits a systematic overestimation error averaging 0.77 m across the scene.

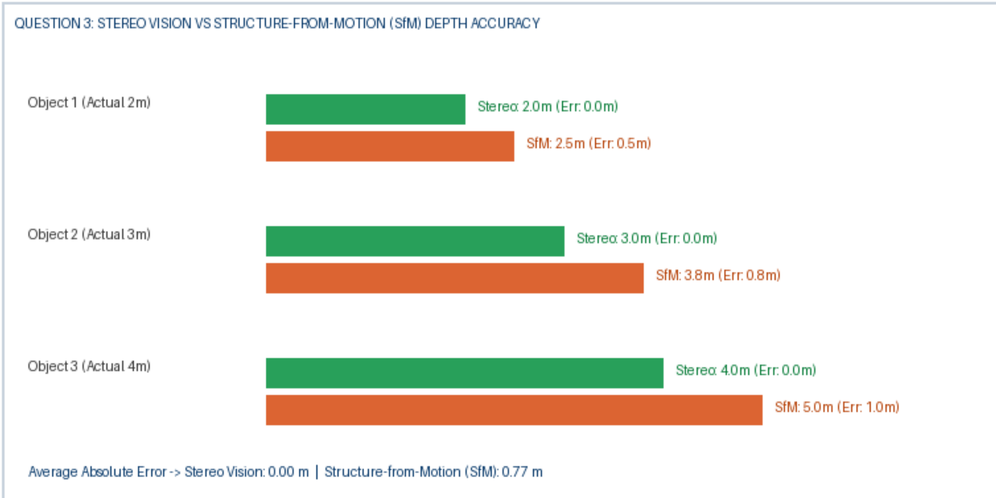

Figure 3: Absolute depth estimation error visual comparison between Stereo Vision and Structure-from-Motion.